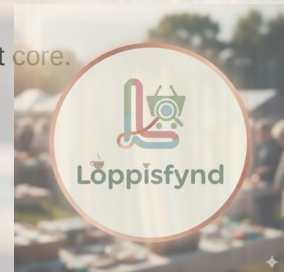


LoppisFynd

Comprehensive review + improved roadmap + UI/UX asset integration

Focus: remove first-run model download, ship cloud-first AI, keep offline-first core.

2026-02-20



1) Executive summary

This document reviews the current LoppisFynd codebase and the provided UI/UX research pack, then proposes an updated roadmap that keeps the Nature Distilled visual direction while removing the first-run on-device model download as a blocker.

What is already working well

- Offline-first architecture (Drift DAOs as source of truth) with clear service boundaries (AI / market / sync).
- Good delivery hygiene: CI runs format/analyze/custom_lint/tests and builds release bundles for staging + prod.
- Design tokens exist in code (colors/spacing/radius/blur/shadows/typography) and shared “glass” primitives are reusable.
- Custom lints enforce “no ad-hoc design constants” and “no raw UI strings”, which prevents UI drift over time.

Top risks / blockers to address next

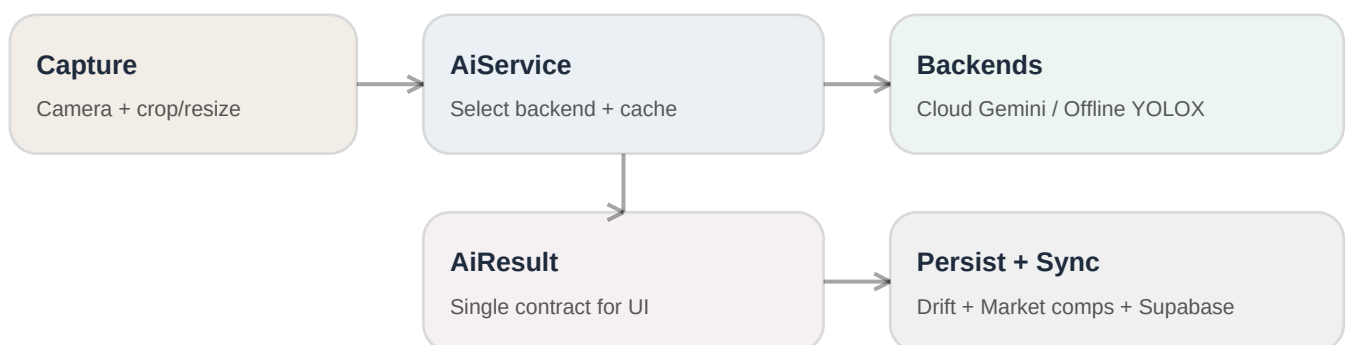
- First-run model download (Gemma) is still wired into onboarding + dashboard; this conflicts with the “download-free first run” goal.
- Cloud AI needs a secure proxy (do not call Gemini directly from the client) plus quotas, auth, and a kill-switch.
- App icon is still the default Flutter mark; branding needs a production-ready icon + simplified logo mark.
- Dark mode is not implemented yet (only light + high-contrast); tokens need light/dark ramps to scale UI work.

2) Codebase review (repo)

Repo snapshot: 140 Dart files under lib/, 40 Dart tests, 38 planning docs, plus Supabase edge functions and migrations.

Architecture (high-level)

The app is feature-first with a shared core/ platform layer. Drift provides offline persistence; Riverpod is used for DI and reactive state; services handle AI inference, market price comps, and optional Supabase cloud sync. This is a strong base for a cloud-first AI pivot because you can swap implementations behind a single service endpoint.



AI implementation (current vs target)

Today, AI inference is isolate-based and uses flutter_gemma with a downloadable model managed by ModelInstallController/ModelManager. To meet the adoption goal, the recommended target is: Cloud-first identification (via a secure proxy) + a small optional offline fallback (e.g., YOLOX) - and no mandatory multi-GB downloads on first run.

Dimension	Current (Gemma-era)	Target (Roadmap v3)
First run	User sees consent + can be blocked by model download	No downloads. Cloud identify works immediately; offline is optional.
Privacy & keys	No cloud calls, but large local footprint	Gemini behind secure proxy (no client key exposure), crop-only uploads.
Failure mode	Disk space / slow download / install issues	Network timeouts degrade gracefully: retry or offline quick ID.
Cost/perf	Device storage + local compute	Cloud costs controlled via quotas + image size caps; cache results in Drift.

Concrete refactors (low-risk, high impact)

- Introduce an AiBackend interface and AiService router (cloud/offline/auto) so UI code calls one place.
- Define a single AiResult contract (JSON schema) used by all backends and persisted in Drift.
- Replace Gemma consent/download surfaces with an “AI status” card (Cloud / Offline / Disabled) + settings toggles.
- Reuse the existing Supabase edge-function pattern (used for Tradera proxy) for an /ai/identify proxy.

3) Improved roadmap (2026–2027)

This version merges the repo’s Nature Distilled UI work with the updated “download-free first run” AI direction. Milestones are designed to be shippable, measurable, and reversible (feature flags).

When	Milestone	Shippable outcome (Definition of Done)
Q1 2026	M1 — Cloud-first AI, download-free first run	Fresh install is usable immediately. Identify uses cloud by default. No model download required. Kill-switch + metrics in place.
Q2 2026	M2 — Lightweight offline fallback	Offline “quick ID” works in airplane mode using a small model (<10MB) and returns bounding boxes in AiResult.
Q2–Q3 2026	M3 — Dependency + platform modernization	Flutter + core packages upgraded incrementally; builds green; no deprecated APIs; CI stable.
Q3 2026	M4 — Dark mode	Token-driven dark theme across all screens; contrast checks; goldens updated.
Q3–Q4 2026	M5 — Responsive layouts	Breakpoints + intentional tablet/foldable layouts for key screens; no overflow.
Q4 2026	M6 — Token formalization	Tokens in a single JSON source of truth (W3C style), exported to Dart and documented.
Q1–Q2 2027	M7 — Future-proofing	Small, measured adoption of 2027 UI/UX trends; performance profiling; decision log.

Critical path (next 3 sprints)

- Sprint 0 (1–2 days): analytics + Sentry breadcrumbs for onboarding + identify; add kill-switch flags.
- Sprint 1: AiResult contract + AiBackend/AiService abstraction; wire UI to use the abstraction (no behavior change yet).
- Sprint 2: Cloud Gemini proxy + app client integration; remove any Gemma auto-download endpoints; ship settings toggles.
- Sprint 3: integration tests + staged rollout; then start YOLOX offline fallback.

Mapping to the existing repo roadmap

The repo’s .planning roadmap is organized as: (1) tokens/primitives, (2) capsule navigation shell, (3) onboarding/auth + model download, (4) core screens + goldens. Phases 1–2 are already aligned with Nature Distilled and should be kept. Phase 3 should be rewritten: replace “Gemma download” with “Cloud AI status + optional offline fallback”.

4) UI/UX pack integration (images + research)

The provided UI/UX pack includes token specs (color/typography/spacing), component guidance (glass + textures), and per-screen layout specs. The codebase already has a strong token/primitives baseline; the main work is aligning token ramps (especially dark mode) and adding texture assets in a performance-safe way.

Pack intent	Current code (AppColors)	Recommendation
Warm neutral background + surface	cloudDancer / clay	Keep as v1, but add neutral ramps (neutral0–neutral4) so borders/elevation scale.
Ink ramp for text	deepSapphire + alpha variants	Add ink0/ink1/ink2 tokens and map typography to semantic roles.
Terracotta brand ramp	terracottaClay + copperOak	Add terracotta ramp (tint→pressed) and use for CTA/selected states.
Dark mode	Not implemented	Implement dark palette + glass rules; add golden tests for core components.

Images: what is safe to ship vs “inspiration only”

Ship-ready assets are textures and the provided logo image. Two reference photos are watermarked (ref1.jpg, ref2.jpg) and should NOT be shipped; keep them as moodboard references only.

File	Type	Suggested new name	Use in app (where)
generated_image_d8a4...png	Texture (paper tile)	Assets/images/textures/paper_wor_n_tile.jpg	Splash/onboarding/empty-state backgrounds (Stack under glass).
furniture_items.png	Texture (linen tile)	Assets/images/textures/linen_tile.jpg	Cards & sheets overlay (10–20% opacity).
vintage_sweaters.png	Texture (wood tile)	Assets/images/textures/wood_tile.jpg	Detail headers, accent strips, subtle overlay.
generated_image_8380...png	Texture (patina coin)	Assets/images/textures/patina_circ_le.jpg	Icon backplates / accent badges (e.g., profit tag).
generated_image_cc74...png	Hero photo	Assets/images/photos/flea_market_1.jpg	Onboarding hero, marketing, optional dashboard hero (blurred).
generated_image_9014...png	Hero photo	Assets/images/photos/flea_market_2.jpg	Onboarding hero variant; app store screenshots.
loppis_background.png	Hero texture	Assets/images/photos/linen_plate.jpg	Splash/empty states; background behind AtmosphericBackground.
logo.jpg / Assets/unnamed.jpg	Logo concept	Assets/brand/logo_lockup.jpg	About screen, onboarding, marketing. (Do not use as app icon.)
ref1.jpg, ref2.jpg	Watermarked refs	(do not ship)	Moodboard only. Replace with licensed/owned imagery.

Ship-ready thumbnails (from the pack)



Paper tile
generated_image_d8a4ceab-d797-4adf-9f76-21bcb62adc8a.png



Linen tile
furniture_items.png



Wood tile
vintage_sweaters.png



Patina accent
generated_image_8380e2e4-7692-41a6-961f-fe0aa4e6e693.png

5) Where to wire UI assets in the Flutter code

Below is a pragmatic “what to change where” list for adding the pack’s textures and replacing the current logo-motif usage.

Goal	File(s) to edit	Change
Add texture overlay behind the existing AtmosphericBackground	lib/shared/widgets/atmospheric_background.dart	Add an optional DecorationImage overlay (low opacity) before the painter track. Gate behind a setting/flag.
Use paper/linen textures for onboarding & empty states	lib/features/onboarding/onboarding_screen.dart; lib/shared/widgets/empty_state.dart	Wrap layouts in a Stack: texture image -> tinted fill -> glass content.
Replace LogoMotifOverlay image with a proper “mark-only” asset	lib/shared/widgets/logo_motif_overlay.dart	Swap AssetImage from Assets/unnamed.jpg to Assets/brand/mark.png (a simplified icon).
Remove Gemma download UI and replace with “AI Status: Cloud/Offline”	lib/features/dashboard/dashboard_screen.dart; lib/features/model_manager/*; lib/features/onboarding/*	Delete or repurpose ModelPreflightCard. Add a status card that reflects AiService backend state.
Add AI settings toggles and enforce them	lib/features/settings/settings_screen.dart + app settings DAO	Implement: “Cloud AI”, “Send crops only”, “Fetch sold comps”, “Offline quick ID”.

Recommended repo asset structure

Move from single-file assets to folder-based assets so you can add more textures without touching pubspec each time:

```
Assets/
brand/
logo_lockup.jpg
mark.png
images/
photos/
flea_market_1.jpg
flea_market_2.jpg
textures/
paper_worn_tile.jpg
linen_tile.jpg
wood_tile.jpg
patina_circle.jpg
```

6) Logo & app icon recommendations

Current situation: the app uses a photo-based circular logo (Assets/unnamed.jpg) for motif overlays, and the actual app icon is still the default Flutter mark. Recommendation: keep the current lockup for marketing/about screens, but create a simplified “mark-only” icon for the app launcher and for subtle motif tiling.

Current lockup (good for About / marketing)



Use cases: About screen, onboarding header, social sharing card. Avoid as launcher icon (too much detail at 48–96px).

Suggested launcher icon (concept)

A launcher icon should read at small sizes: one shape, one idea. Below is a simple vector concept aligned with the palette: terracotta ring + magnifier dot + “L” path. (This is a direction, not a final brand asset.)



Production checklist for branding

- Create mark-only SVG (and export PNGs) for launcher icons (Android mipmaps + iOS app icon set).
- Create monochrome mark for in-app motif overlay (works with tint + alpha).
- Keep the full lockup image for About/marketing; avoid tiling a photo logo in UI for performance.
- Verify license provenance for any third-party photos; remove watermarked refs from shipping assets.

Appendix: Key references (files)

UI/UX research pack (provided)

Key system specs: color_palette_tokens.md, typography_tokens.md, spacing_elevation_tokens.md, glassmorphism_components.md, textures.md. Per-screen specs exist for login, onboarding, dashboard, scanner, history, drafts, item detail, hauls, settings, privacy, sync status, and account deletion.

File	What it covers
account_deletion_screen.md	Per-screen layout + components + states
auth_gate_screen.md	Per-screen layout + components + states
color_palette_tokens.md	Design tokens
color_trends_2026.md	Reference
color_trends_behance_2026.md	Reference
dashboard_screen.md	Per-screen layout + components + states
dashboard_screen_research_2026.md	Trend scan / rationale
deep_link_gate_screen.md	Per-screen layout + components + states
draft_editor_screen.md	Per-screen layout + components + states
drafts_screen.md	Per-screen layout + components + states
glassmorphism_components.md	Glass component rules
glassmorphism_research_2026.md	Trend scan / rationale
haul_screen.md	Per-screen layout + components + states
haul_summary_screen.md	Per-screen layout + components + states
history_screen.md	Per-screen layout + components + states
item_cards.md	Reference
item_cards_research_2026.md	Trend scan / rationale
item_detail_screen.md	Per-screen layout + components + states
login_screen.md	Per-screen layout + components + states
model_manager_screen.md	Per-screen layout + components + states
... (+15 more)	

Repo files to touch for the cloud-first AI pivot (starting points)

Area	Files
AI model download + consent	lib/services/ai/model_install_controller.dart; lib/features/onboarding/onboarding_screen.dart
Gemma backend	lib/services/ai/inference/flutter_gemma_backend.dart; lib/services/ai/inference/inference_isolate_service.dart

Dashboard model preflight card	lib/features/dashboard/dashboard_screen.dart
Settings toggles	lib/features/settings/settings_screen.dart; lib/core/database/daos/app_settings_dao.dart
Edge function pattern	supabase/functions/tradera-proxy (use as template for /ai/identify)
Feature flags	lib/core/config/feature_flags.dart

Note: this PDF intentionally focuses on “what to ship next” rather than repeating every planning document. The codebase already contains a lot of good planning detail; the main delta is replacing the Gemma download path with a cloud-first AI path while keeping the offline-first UX intact.

12) Tech stack addendum (2027-2028 recommendations + execution-ready)

Based on **tech_stack_ui_recommendation_2027_2028.md** (dated 2026-02-20) and a dependency snapshot of the uploaded repo.

Core decisions (keep these stable)

- Stay on Flutter: the app is already cross-platform, well-structured, and the rewrite cost to native would be high with limited upside.
- Cloud-first AI: Gemini (via a secure proxy) becomes the default identification path; offline-first persistence remains unchanged.
- Offline fallback: use YOLOX as a lightweight, commercial-friendly detector (Apache License 2.0) for airplane mode or when cloud is disabled.
- UI/UX evolution: keep the current “Nature Distilled” look, then add dark-mode + token formalization for 2027-2028 readiness.

Why YOLOX (not YOLO11/YOLOv8)

YOLO11/YOLOv8 are AGPL-3.0, which is generally incompatible with distributing a closed-source mobile app without a commercial license. YOLOX is Apache License 2.0 and is therefore the safer default for a production release.

First-run settings toggles (recommended defaults)

- **Send crops for cloud identification** (default ON)
- **Fetch sold-price comps** (default ON)
- **Offline quick ID when no network** (default ON) - routes to YOLOX fallback automatically

Notes on Gemma

Gemma should not block onboarding. If you keep it at all, treat it as a legacy, opt-in backend hidden behind an advanced setting.

13) Dependency modernization - current vs latest

Goal: run on the latest stable toolchain and packages where feasible, while keeping builds green (one big bump per PR).

Snapshot date: 2026-02-20.

Component	Current (repo)	Latest stable	Action / Notes
Flutter SDK	>=3.38.4	3.41.x	Pin via FVM/CI; target latest stable
Dart SDK	^3.10.8	3.11.x	Raise SDK constraint to match Flutter's Dart
Riverpod	2.6.1	3.2.1	Major upgrade (2.x → 3.x) - plan migration
Drift	2.31.0	2.31.0	Already current
drift_dev	2.31.0	2.31.0	Already current
drift_flutter	0.2.8	0.2.8	Already current
Camera	0.11.3+1	0.11.4	Safe patch upgrade
Workmanager	0.9.0+3	0.9.0+3	Already current
ML Kit barcode	0.14.2	0.14.2	Already current
Sentry	9.13.0	9.14.0	Safe minor/patch upgrade
Supabase	2.12.0	2.12.0	Already current
Notifications	20.1.0	20.1.0	Already current
Permissions	12.0.1	12.0.1	Already current
Package info	9.0.0	9.0.0	Already current
HTTP	1.6.0	1.6.0	Already current
Google Fonts	6.3.3	8.0.2	Major upgrade; check typography rendering
SQLite native libs	0.5.41	0.6.0+eol	0.6.0+eol is for sqlite3 3.x; don't bump until drift/sqlite3 migrate
Gemma	0.12.4	Remove	Remove from first run; keep only if legacy opt-in
YOLOX plugin	Not present	1.1.1	Add as offline fallback (Apache 2.0 friendly)

Interpretation: Drift / Supabase / Workmanager / MLKit are already on current stable. The main immediate wins are: pin Flutter to 3.41.x, bump camera and Sentry, remove Gemma, and plan Riverpod + Google Fonts majors carefully.

14) Upgrade plan - keep it running on latest stable

This is the safest sequencing to modernize without stalling feature delivery.

Phase A (1-2 PRs): Pin toolchain

- Pin Flutter to stable 3.41.x in CI (and ideally via FVM locally).
- Raise pubspec SDK constraints to match the pinned Flutter (Dart 3.11.x+).
- Run a clean build + tests; fix any analyzer or gradle/xcode warnings.

Phase B (2-4 PRs): Safe package bumps

- camera: 0.11.3+1 → 0.11.4.
- sentry_flutter: 9.13.0 → 9.14.0.
- Keep drift/drift_dev/workmanager/mlkit as-is (already current).

Phase C (feature PRs): AI stack swap (critical path)

- Remove flutter_gemma from first-run and delete any automatic model download triggers.
- Ship cloud Gemini backend behind a secure proxy (Edge Function) and add kill-switch + quotas.
- Add YOLOX offline fallback using flutter_yolo_open_kit (Apache 2.0) and surface evidence boxes in the UI.

Phase D (migration PRs): Major upgrades (as capacity allows)

- Riverpod 2.x → 3.x: do a dedicated migration PR with zero feature changes.
- google_fonts 6.x → 8.x: validate typography metrics and any runtime font fetching behavior.
- sqlite3 2.x → 3.x: only if drift ecosystem supports it; otherwise hold.

References (public)

Flutter release notes (docs.flutter.dev); Dart SDK archive (dart.dev); pub.dev package pages for dependencies in the table; YOLOX LICENSE (Apache License 2.0) on raw.githubusercontent.com.

15) Implementation guide (execution-ready)

How to execute the roadmap without stalling (cloud-first AI + Nature Distilled UI).

Guiding rules

- PR-sized work (<= 1-2 days). One primary change per PR.
- Feature flags for risky behavior (cloud AI kill-switch; staged rollout).
- Prefer transforms + opacity for motion; do not animate blur radius or heavy shadow stacks.
- Accessibility: support reduced motion (route slide -> 0, scanner laser static).

W1 - AI & ML (cloud-first + offline fallback)

- Define AiResult JSON contract and persist it in Drift (single UI contract).
- Introduce AiBackend + AiService router (cloud/offline/auto) so UI calls one endpoint.
- Implement Supabase Edge Function: POST /ai/identify (Gemini proxy, quotas, auth, logging).
- Client: crop + compress images, call proxy, cache results; degrade gracefully on timeouts.
- Offline: integrate YOLOX fallback (<10MB target) and return evidence boxes in AiResult.
- Remove Gemma codepaths + any auto-download triggers; replace download UI with AI status.

W3 - UI/UX (Nature Distilled, tokens-first)

- Lock palette ramps + typography roles (Outfit for UI/body; Space Grotesk for numbers).
- Wire tokens into ThemeData (light + dark); add goldens for core components.
- Component pass: GlassSurface/Card/Button/Input/Chip + layered shadows + grain overlay.
- Screen rollout order: Dashboard+Scanner -> Item cards/detail -> Drafts -> History/Hauls -> Settings/Privacy -> Login/Onboarding.

W4 - Quality, release, observability

- Analytics events: first_run, identify started/success/fail, backend selected.
- Sentry breadcrumbs for onboarding + capture + identify failures.
- Integration tests: fresh install (no downloads), capture->identify cloud, capture->identify offline.

16) Phase list (PR-sized deliverables)

Each phase is intended to be shippable; split into PRs as needed.

Phase	Name	DoD / Outcome	Touchpoints
Phase 0	Instrumentation + flags	Events + Sentry + kill-switch; baseline metrics	analyticsProvider, feature_flags, onboarding/camera flows
Phase 1	AI contract + routing	AiResult model + AiBackend/AiService/core/ai/*, Drift behavior change yet	schema/cache
Phase 2	Cloud identify via proxy	Supabase /ai/identify + client integration; cache; graceful failure	supabase/functions/ai-identify, client service
Phase 3	Remove Gemma surfaces	Delete/disable Gemma + model downloading; show AI status	onboarding, dashboard preflight, settings AI
Phase 4	YOLOX offline fallback	Offline quick ID in airplane mode; evidence boxes in AiResult	offline backend + mapping
Phase 5	Tokens + dark mode	Token-driven ThemeData; core component goldens pass	app_tokens + component widgets
Phase 6	Screen rollout	Per-screen updates in priority order; raw hex or ad-hoc constants	feature screens + goldens
Phase 7	Motion pass	Reduced motion policy + pressable consolidation + 2-3 continuity animations	AppMotion, SpringRoute, sidebar overlay, pressables